

VAMSI KRISHNA SAI THOTA

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SUMMARY

I'm an analytical and detail-oriented data professional with around **5+ years** of experience in Business Intelligence, data analysis, reporting and automation across operational and customer experience teams. I am skilled in **Tableau (Tableau Desktop Specialist), Power BI, SQL, Excel (advanced), and data storytelling, Microsoft Fabric, Data Factory, Databricks** with a proven track record of maintaining and developing business-critical reports, automating workflows, and driving data-informed decision-making. Adept at collaborating with leadership and cross-functional teams to deliver accurate insights and operational excellence.

SKILLS

- **Visualization and Reporting:** Python, R, Tableau, Power Bi, Dashboard design, report automation, DAX
- **Data Engineering & Cloud:** Microsoft Azure – Fabric, AWS – S3, SQL, Python, R, Kafka, Data Factory, Databricks, Data Lake, SAP
- **ETL & Data Warehousing:** Microsoft Fabric, Azure Data Factory, Informatica, ETL/ELT, Data Warehouses and Lakehouse
- **Databases:** SQL, SAS, MySQL, R, PostgreSQL, Snowflake, Microsoft SQL Management Studio
- **Advanced Excel:** Pivot tables, Nested formulas, VBA Macros, conditional formatting, calculated fields
- **Software Development:** SDLC, Agile Methodologies, Project Management (JIRA, Confluence), UI/UX Design basics

EXPERIENCE

CVS HEALTH, USA

02/2025 – Present

DATA ANALYTICS ENGINEER

- Designed and developed **interactive Power BI dashboards** in **Azure Fabric** to provide executives and business units with actionable insights on claims, underwriting, customer retention, and policy performance. Implemented row-level security in Power BI Service, controlling data visibility for 95+ users and supporting HIPAA compliance across analytics assets using Git-managed publishing pipelines.
- Automated **ETL workflows** in **Azure Fabric** and Databricks to ingest, transform, and validate large volumes of customer and claims data from multiple sources, reducing manual intervention and ensuring data quality. Automated monthly data pipelines in Azure Data Factory and Snowflake, cutting manual processing by 80% while increasing report refresh rates from weekly to daily without IT dependency.
- Performed **advanced SQL data modeling** (joins, CTEs, window functions) to support reporting and predictive analytics initiatives across claims, fraud detection, and marketing operations. Delivered churn analysis and CLV modeling in Python and SAS, identifying high-risk patient segments and increasing retention efforts that led to a 13% drop in Q2 customer attrition.
- Partnered with teams to identify suspicious claims patterns, implement fraud detection dashboards, and improve early identification of high-risk activities. Worked closely with **cross-functional teams** (actuarial, finance, claims operations, IT) to translate business requirements into scalable BI solutions. Actively contributed to an **Agile/Scrum environment**, managing deliverables via JIRA and Confluence, ensuring timely delivery of data-driven solutions.

DEVELOPMENT BANK OF SINGAPORE (DBS TECH), INDIA

06/2022 to 07/2023

DATA ANALYST

- I worked on the **credit card fraud analysis** project within the **Risk & Management** team by migrating **SAP data (~ 1.3 million records)** to **Azure using ADF**, Blob Storage, Snowflake and **ETL pipelines**. Applied advanced SQL for data modeling. Using **POWERBI**, I developed dynamic dashboards with slicers and filters to analyze churn, retention, and fraud trends. The project enabled real-time insights, automated reporting, and improved fraud detection and customer targeting.
- Spearheaded the migration of customers and transactional data from SAP systems to Azure cloud platforms, enabling advanced fraud detection analytics for credit card transactions. Designed and implemented scalable ETL workflows using **Azure Data Factory (ADF)**, including **copy activities, SQL scripts**, and scheduled **semantic model refreshes**, significantly reducing latency in fraud data processing pipelines.
- Built **semantic layers** and **data models** to prepare fraud-risk data for **POWERBI** dashboards. Designed and developed **dynamic, auto-refreshing dashboards** with **advanced measures**, custom filters, slicers, calculated columns/tables, and **append/merge queries**. Visualizations enabled stakeholders to monitor fraud metrics, evaluate ad-hoc marketing impact, and track churn/retention trends.
- Delivered insights through compelling **data storytelling**, identifying actionable opportunities to enhance customer trust, reduced false positives in fraud flags, and improve satisfaction through **targeted engagement strategies**. Developed robust **data pipelines between Azure Blob Storage and Snowflake** to ingest structured and semi-structured data. Applied **data profiling, transformation, validation, and semantic checks**, adhering to strict **MDM and data governance principles** to ensure clean, analysis-ready data for the Risk team.
- Supported multiple global clients across supply chain, transportation, and marketing industries, working closely with cross-functional teams to deliver Data Analytics using **Power BI** and data-driven solutions tailored to business needs. Built **end-to-end ETL pipelines using Azure Data Factory, handling diverse data sources like Excel, MySQL, XML, and REST APIs**; collaborated with data engineers to streamline ingestion from Azure Blob Storage into SQL Server and Snowflake
- Designed and deployed dynamic **Power BI dashboards** on customer acquisition cost, marketing expenses, freight analysis, and email usage, enhancing executive decision-making. Conducted **freight gap analysis in Power BI using advanced DAX measures** by querying and merging transportation and delivery data, using **SQL nested queries and functions** to flag missing freight info and visualize KPIs like affected deliveries, involved DCs, and customer groups.

WEBSITES FACTORY, INDIA

ANALYST - POWER BI DEVELOPER

01/2021 to 04/2022

- Gathered and interpreted business requirements from customers and stakeholders to identify key performance indicators, define analytical goals, and guide the development of data-driven solutions supporting both business operations and marketing strategies. Performed end-to-end data collection, cleansing, and preprocessing using **SQL** for structured data and **R programming** for advanced statistical analysis, ensuring accuracy and consistency across multiple reporting layers.
- Utilized **SAP** systems to extract customer transactions and interaction data, enabling the generation of reliable, real-time reports. Transformed and merged complex datasets to create a unified customer view for deeper behavioral and trend analysis.
- Designed and developed **interactive Power BI dashboards** to visualize customer satisfaction metrics, purchasing patterns, and marketing engagement outcomes. Applied **DAX measures** and advanced data modeling techniques to build insightful visualizations that empowered cross-functional teams to make strategic decisions.
- Built **email exchange analytics (~ 100K records) dashboard in Power BI** with time-based slicers, summarizing usage trends by department and user groups, supporting IT audit and licensing decisions. Automated semantic model refreshes, database updates, and report scheduling using Azure and Power BI services, reducing manual effort and improving report accuracy. Conducted **A/B testing** on marketing and product performance metrics to compare customer engagement outcomes, optimize campaign strategies, and support data-driven decision-making using Power BI and SQL. Used Confluence for documentation and GIT for version control and change management.
- Developed marketing insights based on segmentation analysis, customer lifetime value modeling, and churn prediction models in **R**, significantly improving campaign targeting and ROI. Worked in an **Agile/Scrum** environment with active participation in sprint planning.

EDUCATION

MASTER OF SCIENCE - COMPUTER SCIENCE (GPA:3.8/4)

08/2023 -12/2024

Wright State University (WSU), Fairborn, Ohio

Algorithm Design and Analysis, **Information Retrieval**, **Data Analysis**, **Machine Learning**, **Advanced AI**, Advanced Computer Networks, Cryptography and Data Security, Distributed Computing, Cyber Network Security

BACHELOR OF TECHNOLOGY - INFORMATION TECHNOLOGY (GPA:8.5/10)

06/2018 - 05/2022

Malla Reddy College of Engineering & Technology (MRCET), Hyderabad, India

Data Visualization, R Programming, **Statistics**, **Discrete Mathematics**, **Relational Database Management**, Data Science, Machine Learning, Artificial Intelligence, Data Structures, Cloud Computing, Python Programming, C# Programming, XML Scripting

ACADEMIC WORK

LEGO Themed Interactive Dashboard, Individual Spring 2025

This project involved designing a dynamic Power BI dashboard to explore LEGO product data, featuring over 4,000 records. I implemented numeric range slicers, image-based tool tips, bookmarks for resetting filters, and decomposition trees for enhanced navigation. By building a clean data model and developing DAX-based KPI cards, I enabled users to analyze sets by theme, price, and age group, creating an intuitive and visually engaging exploration experience

Distributed Deep Learning with Azure Fall 2023

Enhanced ML models (CNN, LSTM, Bi-RNN) using Azure Cloud (Azure AI/ML) and parallel computing, improving system performance by 20%. Applied dynamic resource allocation and edge computing for optimized scalability

ML Algorithms for Disease Prediction – Web-Based Application Spring 2022

I created a web-based disease prediction application using Python (Flask) and integrated four machine learning algorithms—Naïve Bayes, Random Forest, Decision Tree, and Support Vector Machine (SVM)—to predict diseases based on user-input symptoms. After training and validating models, I evaluated them using accuracy, precision, recall, and F1 score metrics. The project involved building a user-friendly web interface to allow real-time predictions, aimed at supporting early disease detection for patients and healthcare providers.

LEADERSHIP EXPERIENCE

Graduate Teaching Assistant – CS1160 Introduction to Computer Programming with Python - Fall 2024

- Supported 80 undergraduate students in mastering Python programming fundamentals by leading lab sessions, assisting with coursework, and providing academic guidance during weekly office hours. Graded assignments and exams, delivering constructive feedback to help students improve their programming skills.

CERTIFICATIONS

- **Certified Tableau Developer – Tableau Desktop Specialist**
- Certification - **Business Analysis Basics**
- Certification – **Power BI: Dashboards for Beginners**
- Certification – **9.4 SAS Programming Essentials**
- Coursera & IBM certifications – Python, Excel, SQL, Database Management, Snowflake, Power BI, R, Azure Data Factory and Databricks.